

Option #1

Title: Adaptation of Vision Foundation Models for Knee Osteoarthritis Classification

Due: 05/01/2026; 11:59pm EDT

Objective:

In this project, you will develop and evaluate an AI system to classify **knee osteoarthritis (OA)** vs. **non-OA** from radiographs using **pretrained vision foundation models** such as:

- DINOv2
- CLIP
- MedCLIP (*optional / bonus*)

You will systematically analyze how **data availability impacts performance** by training models under:

- **20% of training data (low-data regime)**
- **50% of training data (moderate regime)**
- **100% of training data (full-data regime)**

This mirrors real-world clinical AI challenges where labeled data is limited

Problem Formulation:

You will implement a **binary classification task**:

- **Label 0:** $KL < 2 \rightarrow$ No OA
- **Label 1:** $KL \geq 2 \rightarrow$ OA

This is a clinically meaningful threshold used in practice

Dataset:

We have made a dataset ready for you.

- Knee radiographs (AP view)
- Associated labels as Knee OA versus No-Knee OA

You must:

- Split data into **train / validation / test**
- Ensure **no subject-level leakage**

Preprocessing:

- Resize images to **224×224**
- Normalize based on model requirements
- (Optional Bonus) ROI extraction if available

Model Implementation:

You must implement **at least ONE** of the following:

Option A (Recommended)

- DINOv2 (ViT backbone)

Option B

- CLIP (ViT-B/32)

Option C (Advanced / Bonus)

- MedCLIP

Classifier Head:

Use a simple **MLP classifier**: Linear → ReLU → Dropout → Linear → Softmax

Training Setup:

- Loss: **Cross-Entropy**
- Optimizer: **AdamW**
- Batch size: 16 (recommended)
- Epochs: 10–20
- Early stopping based on validation

Evaluation Metrics:

You must report:

- Accuracy
- Precision
- Recall
- F1-score
- **Macro-F1 (important)**
- ROC-AUC

Analysis:

You must answer:

1. Data Efficiency

- How does performance change from 20% → 100%?

2. Model Behavior

- Which model performs best and why?

3. Adaptation Strategy

- Does fine-tuning always help?

4. Stability

- Which setup is most consistent?

Deliverables:

1. Code (Python only – GitHub Repository)

- Well-documented
- Reproducible
- Invite MariaGOY as a collaborator to your GitHub repository.

2. Report (4–6 pages)

Must include:

- Introduction
- Methods
- Experimental setup
- Results (tables + plots)
- Discussion

3. Presentation (5–7 minutes)

Expected Learning Outcomes:

By the end, students will:

- Understand **foundation models in medical imaging**
- Learn **data efficiency challenges**
- Implement **transfer learning in healthcare AI**
- Analyze **model performance beyond accuracy**